

HDOC DAY-40

Name-Priyadarshna

Sapid-590032928

Batch-15

Question: Write a menu-driven C program using user-defined functions to perform string operations on a student's name: combine, find length, copy, compare, reverse, count characters, search a word, and exit.

Aim:

To write a menu-driven C program to perform different string operations on a student's name using user-defined functions.

Header Files:

stdio.h, string.h, ctype.h

Functions used:

strcpy(), strcat(), strlen(), strcmp(), strstr(), isalpha(), isdigit(), isspace(), tolower()

Operations:

1. Combine names
2. Find length
3. Copy name
4. Compare names
5. Reverse name order
6. Count vowels, consonants, digits & spaces
7. Search a word
8. Exit

Algorithm:

1. Start.
2. Enter first, middle and last name.
3. Display the menu.
4. Enter the user's choice.
5. Perform the selected string operation using a function.
6. Display the result.
7. Repeat until **Exit** is selected.
8. Stop.

>C PROGRAM

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
int main(){
```

```
    char f[20], m[20], l[20], full[60];
```

```
    int ch;
```

```
    printf("Enter first middle last name: ");
```

```
    scanf("%s%s%s", f, m, l);
```

```
    strcpy(full, f);
```

```
    strcat(full, " ");
```

```
    strcat(full, m);
```

```
    strcat(full, " ");
```

```
    strcat(full, l);
```

```
    do {
```

```
        printf("\n1.Full Name\n2.Length\n3.Compare\n4.Exit\n");
```

```
        printf("Enter choice: ");
```

```
        scanf("%d", &ch);
```

```
        switch(ch) {
```

```
            case 1: printf("%s", full); break;
```

```
            case 2: printf("%zu", strlen(full)); break;
```

```
            case 3:
```

```
                printf("%s", strcmp(f,l)==0 ? "Equal" : "Different");
```

```
                break;
```

```
        }
```

```
    } while(ch != 4);
```

```
    return 0;
```

```
}
```

>OUTPUT:

Enter first middle last name:

Aman Kumar singh

1. Full Name

2. Length

3. Compare

4. Exit

Aman Kumar Singh

1.Name 2.Length 3.Compare 4.Exit

1

Aman Kumar Singh

1.Name 2.Length 3.Compare 4.Exit

2

15

1.Name 2.Length 3.Compare 4.Exit

3

Different

1.Name 2.Length 3.Compare 4.Exit

4